

# Cressington **208HR**

## High Resolution Sputter Coater



Illustrated: sputter coater 208HR with: rotary-planetary-tilt stage; thickness controller MTM-20

### The main features are:

- Wide Choice of Coating Materials:-  
Magnetron head design and effective gas handling allow a wide choice of target materials (see specification).
- Precision Thickness Control:-  
Thickness optimized to the FE-SEM operating voltage using the MTM-20 high resolution thickness controller.
- Multiple Sample Stage Movements:-  
Separate rotary, planetary and tilting movements allow optimized coating distribution and coverage. (view RPT Stage)
- Variable Chamber Geometry:-  
Chamber geometry is used to adjust deposition rates from 1.0 nm/sec to 0.002nm/sec to optimize structure.
- Wide Range of Operating Pressures:-  
Independent power/pressure adjustment allows operation at argon gas pressure ranges of 0.2 - 0.005 mbar.
- Compact Modern Benchtop Design:-  
Space and energy saving design eliminates need for floor space, water, specialized electrical connections.

### Coating Difficult Samples for FE-SEM

The Cressington 208HR now offers real solutions to the problems encountered when coating difficult samples for FE-SEM. In order to minimize the effects of grain size the 208HR offers a full range of coating materials and gives unprecedented control over thickness and deposition conditions. To minimize charging effects the 208HR stage design and wide range of operating pressures allows precise control of the uniformity and conformity of the coating. The HIGH/LOW chamber configuration allows easy adjustment of working distance. (Compare this picture with the top picture.)



## High resolution sputter coater

|                             |                                                                                                                                                                               |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Chamber size</b>         | 12 150mm Ø, Variable height, 165mm - 250mm                                                                                                                                    |
| <b>Sputter head</b>         | Low voltage planar magnatron<br>Quick target change<br>Wrap-around dark-space shield<br>Shutter for target conditioning                                                       |
| <b>Sputter target</b>       | Cr, Pt/Pd as standard<br>(Ta, Au, Au/Pd, Pt or W optional)                                                                                                                    |
| <b>Sputter supply</b>       | Microprocessor based<br>Safety interlocked<br>Current control independent of vacuum<br>Digitally selectable current (20, 40, 60 or 80mA)                                      |
| <b>Sample table</b>         | Non-repetitive rotary planetary motion with<br>manual tilt 0 - 90°<br>Variable speed rotation<br>Crystal head<br>4 sample holders                                             |
| <b>Analogue metering</b>    | Vacuum, Atm - 0.001mb<br>Current, 0 - 100mA                                                                                                                                   |
| <b>Control method</b>       | Automatic operation of gas purge and leak<br>functions<br>Automatic process sequencing<br>Full manual override<br>Digital timer, 5 - 300 seconds with pause<br>Automatic vent |
| <b>Thickness monitoring</b> | MTM-20 with termination facility                                                                                                                                              |

## Pumping system

|                          |                                                                                                          |
|--------------------------|----------------------------------------------------------------------------------------------------------|
| <b>Rotary Pump</b>       | Turbo-drag/rotary pump combination<br>Optional diaphragm pump instead of rotary pump                     |
| <b>Pumping speed</b>     | 300 litres/min @ 0.1mb                                                                                   |
| <b>Pumpdown time</b>     | 1 min. to $1 \times 10^{-3}$ mb<br>(1.5 min. with diaphragm pump option)                                 |
| <b>Ultimate pressure</b> | $1 \times 10^{-5}$ mb                                                                                    |
| <b>Desktop system</b>    | Rotary pump is mounted on desktop compatible<br>anti-vibration table<br>All metal vacuum coupling system |

## Thickness monitor

|               |                                                                                                                                         |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>MTM-20</b> | Microprocessor based<br>4 digit display with push button zero<br>5 times/sec. display update rate<br>6 MHz crystal with life-time check |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------|

**Thickness range** 0 -999.9nm

**Resolution** Better than 0.1nm

**Density range** 0.50 - 30.00gm/cm<sup>3</sup>

**Tooling factor range** 0.25 – 8.00

**Termination range** 0 -999.9nm

### **Service required**

**Supply** 100 - 120 or 200 - 240 VAC, 50/60Hz  
(specify on order)

**Power** 550 VA max.

**Argon gas** Purity, min. 99.9%  
Pressure, regulated 7 - 8 psi (0.5 - 0.6 bar)  
Hose connection, 6.0mm (¼")

### **System dimensions**

**Size** Width 600mm (23.6"), Depth 600mm (23.6"),  
Height 450mm (17.7")

**Weight** 40 kg